

- večinoma v režimu $v \ll c$ (E so iz atomskega/molekularnega sveta)

↓
mikroskopske prostorske skale

3 glavne značilnosti:

1) E, A, ... se izmenjujejo (med sistemoma) le v kvantih (obrokih)

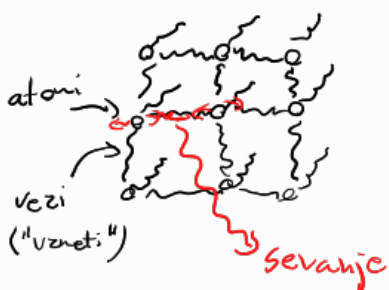
2) V kvantnem svetu se lahko delci donesajo kot valovanje, in valovanje kot delci

3) klasično: $\vec{r}(0), \dot{\vec{r}}(0), F(t \geq 0) \rightarrow \vec{r}(t), \dot{\vec{r}}(t)$ za $t > 0$ po Newtonu

kvantno: lahko napovemo samo verjetnost za izid poskusa

Sevanje črnega telesa

- že pred koncem 19. stol.: prve indikacije za kvantno naravo EM valovanja teles v term. ravn. (temp. T)

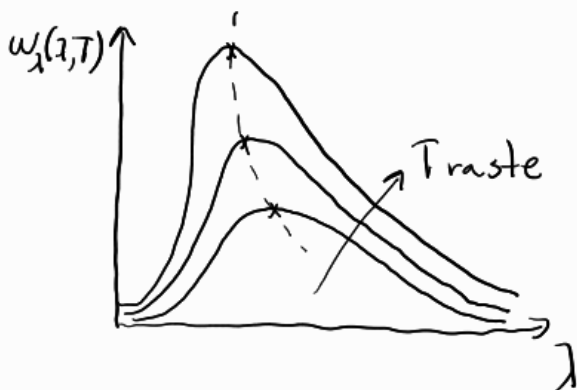


Termično (topl.) sevanje

- seva ga vsako telo pri $T > 0K$

- šele pri $T \approx 700^\circ C$ je tega sevanja dovolj tudi v vidnem delu spektra, da to vidimo kot žarenje

Energ. spekter v merilu val. dolžin



maksimum: $\boxed{\lambda_w = \frac{c_w}{T}}$ Wienov zakon

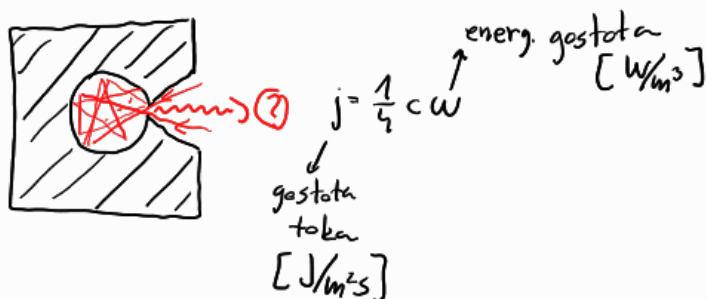
$$c_w = 2,898 \cdot 10^{-3} \text{ mK}$$

$\boxed{j = \sigma T^4}$ Stefanov zakon

Idealno črno telo-telo, ki absorbira vse vpadno sevanje

Kako do spektra?

→ štetje načinov EM valovanja v votlini, aproksimacija:



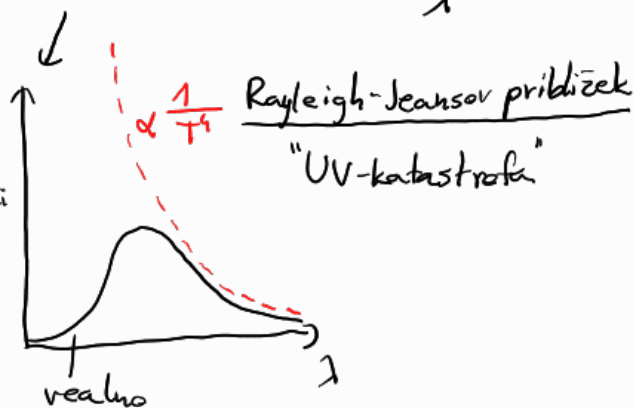
nihajnih načinov (na enoto λ) je neodv. od dolžine resonatorja

$$n(\lambda) = \frac{dn}{d\lambda} = \frac{8\pi}{\lambda^4} \quad (\text{rezultat iz klas. fizike})$$

→ iz stat. fizike: vsakemu nihajnemu načinu ustreza energija $k_B T$

Torej bi klas. pričakovali

$$w_\lambda(\lambda, T) = k_B T \cdot n(\lambda) = \frac{8\pi k_B T}{\lambda^4}$$



→ kako utežiti načine z majhnimi

λ , da bo $\lim_{\lambda \rightarrow 0} w_\lambda = 0$?

PLANCKOV ZAKON (izpeljava)

- osn. ideja: energija nih. načinov je lahko samo cel mnogokratnik neke vrednosti

$$E_n = n \underbrace{\epsilon}_{\text{kvant}} = n h \nu = n h \frac{c}{\lambda} ; n=0,1,2,\dots$$

Planckova konstanta

$$h = 6,6 \cdot 10^{-34} \text{ Js}$$

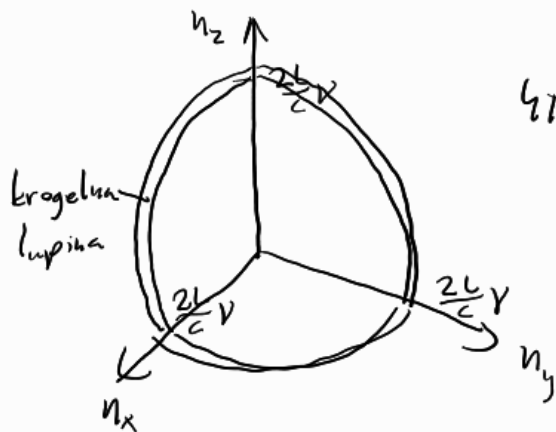
$$\hbar = \frac{h}{2\pi} = 1,05 \cdot 10^{-34} \text{ Js}$$

3 je ključni koraci:

- 1) prešteti # načinov, ki jih pri dani T spravimo v votline z volumenom V
- 2) verjetnost za posamezen način
- 3) povprečna E pri dani T

1) analogija z 1d (struna)

 $L = n \frac{\lambda_n}{2} \rightarrow n = \frac{2L}{\lambda_n} \xrightarrow{v = \lambda f} \sqrt{n_x^2 + n_y^2 + n_z^2} = \frac{2L}{\lambda_n} v_{n_x, n_y, n_z}$



$$4\pi r^2 dr \rightarrow \frac{1}{8} 4\pi \left(\frac{2L}{c}v\right)^2 \frac{2L}{c} dv = \frac{4\pi v^2}{c^3} \underbrace{L^3}_V dv$$

→ pomnožimo še z 2 zaradi polarizacije

$$d(\# \text{ stars}) = \frac{8\pi v^2}{c^3} v dv$$

$$N_\nu = \frac{8\pi \nu^2}{c^3} V$$

$w_y(v, T) = \frac{N_v}{V} \langle \epsilon \rangle$ $\rightarrow$ dobimo z uporabo Boltzmannove porazdelitve:

$$\langle \epsilon \rangle = \frac{\sum_n f(E_n) E_n}{\sum_n f(E_n)}$$

$f(\epsilon)$ - gost. Boltzman. porazd.

$$f(\epsilon) = e^{-\epsilon/k_B T}$$

$$\langle E \rangle = \frac{\sum_{h\nu=0}^{\infty} h\nu \cdot e^{-\frac{h\nu}{k_B T}}}{\sum_{h\nu=0}^{\infty} e^{-\frac{h\nu}{k_B T}}} =$$

$$= \frac{k_B T \sum_n n x e^{-nx}}{\sum e^{-nx}} = k_B T \frac{x e^{-x}}{(1-e^{-x})^2} (1-e^{-x})$$

imеновалек: $z = \sum e^{-tx} = 1 + e^{-x} + e^{-2x} + \dots = \frac{1}{1 - e^{-x}}$

3. tevec: $-x \frac{d^2}{dx^2} = -x \frac{d}{dx} \sum e^{-nx} = x \sum n e^{-nx}$

$$\rightarrow x \int h e^{-hx} = -x \frac{d}{dx} \left(\frac{1}{1-e^{-x}} \right) = \frac{x e^x}{(1-e^{-x})^2}$$

$$\Rightarrow w_\nu(\nu, T) = \frac{8\pi\nu^2}{c^3} \langle E \rangle = \frac{8\pi\nu^2}{c^3} k_B T \frac{x}{e^x - 1} = \boxed{\frac{8\pi h \nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}}$$

$$w_\lambda = w_\nu \left| \frac{d\nu}{d\lambda} \right| = \boxed{\frac{8\pi h c}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1}}$$

$$j = \int_0^\infty \frac{c}{4} w_\lambda d\lambda = \int_0^\infty \frac{c}{4} w_\nu d\nu = \boxed{\sigma T^4}$$

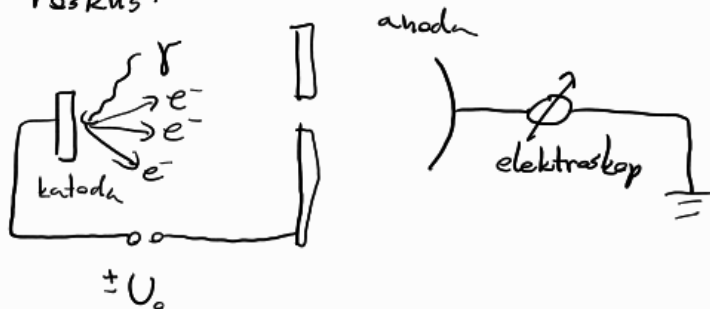
Fotoel. pojav (fotoefekt)

Planck-črno telo seva v kvantih

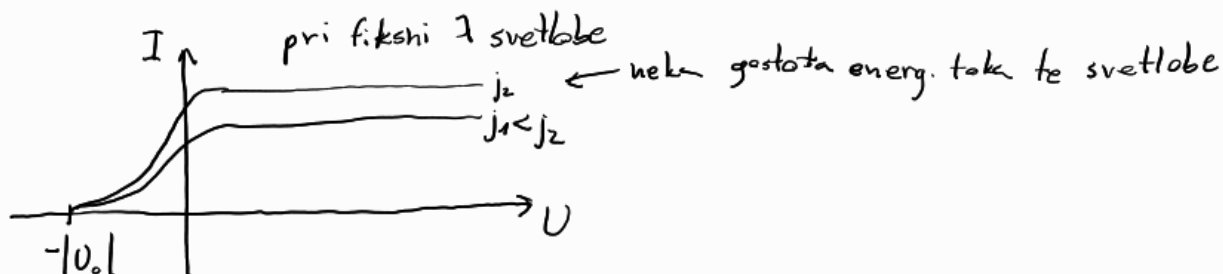
→ Einstein - to je splošna lastnost svetlobe



Poskus:

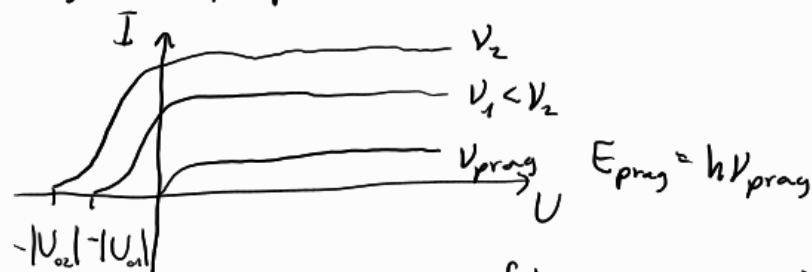


(pospešimo ali
zaviramo e^-)



zastopna napetost → ni odvisna od jakosti → $eU_0 = \left(\frac{1}{2}mv^2\right)_{\max}$
(nič e^- ne pride skozi) → padne svetlobe

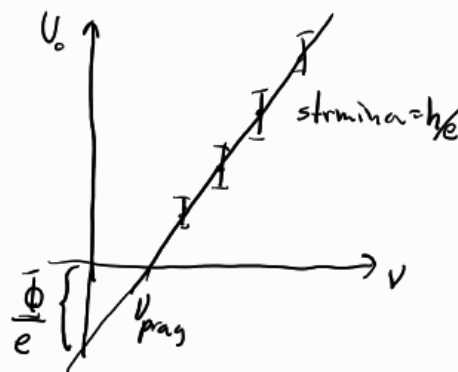
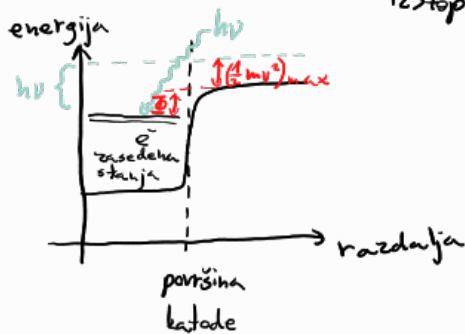
$j = \text{konst.}, \text{ spreminjamo } \lambda \text{ (oz. } \nu \text{):}$



fotoni z energijami $< E_{\text{prag}}$
ne izbijajo nobelih e^-

$$eU_0 = \left(\frac{1}{2}mv^2\right)_{\text{max}} = h\nu - \Phi$$

izstopno delo (potrebno, da spravimo e^- iz kovine)



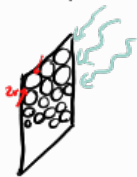
tipični Φ : nekaj eV

Zgled: Katoda K ($\Phi = 2,2 \text{ eV}$)

$$\lambda = 400 \text{ nm}$$

$$j = 10^{-2} \text{ W/m}^2$$

tipični polmer atoma $r \approx 10^{-10} \text{ m}$



energija, ki v času t pade na atom: $E = j\pi r^2 t \frac{1}{e} \Phi$

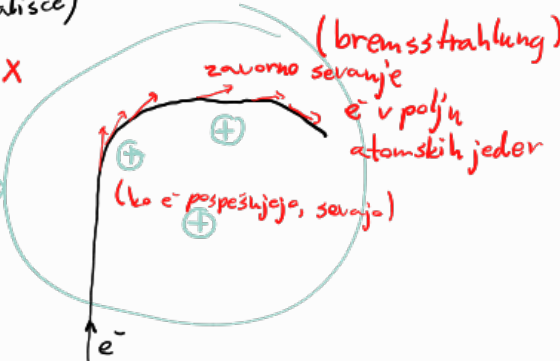
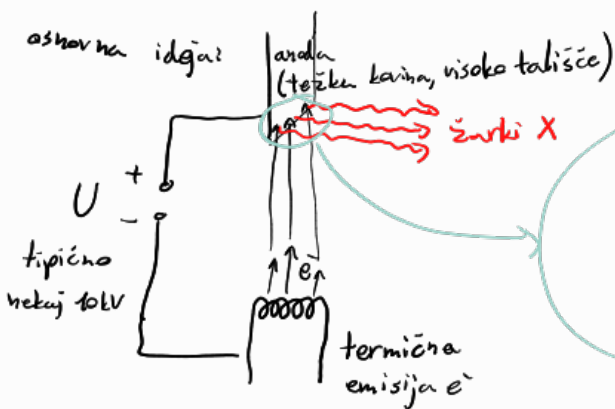
$$t = \frac{\Phi}{j\pi r^2} = 19 \text{ min}$$

toliko bi morali počakati, da bi izbili en sam e^- , v resnici pa se e^- pojavijo takoj

$\Rightarrow$ vsak γ ima dovolj E , da izbije e^-

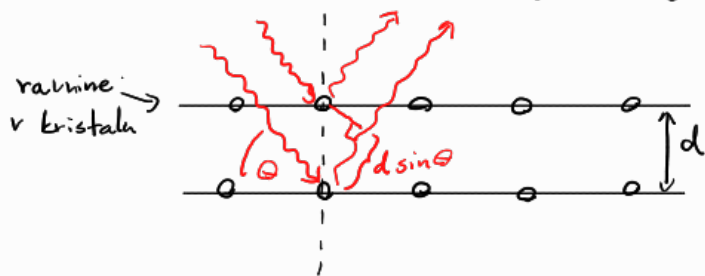
Röntgenovo sevanje (x-rays)

osnovna ideja: anoda (težka kovina, visoko tališče)



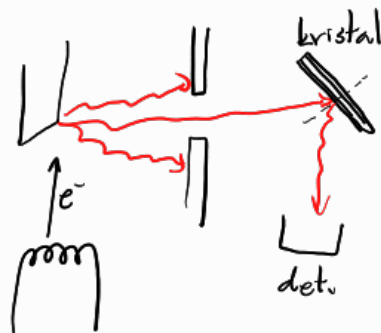
že Röntgen je opazil, da so el. nevtralni (mag. polje jih ne ukloni) in $\lambda \approx 0,1 \text{ nm}$.

Kerko analiziramo RTG sevanje (Bragg)



$$2d \sin \theta = n\lambda$$

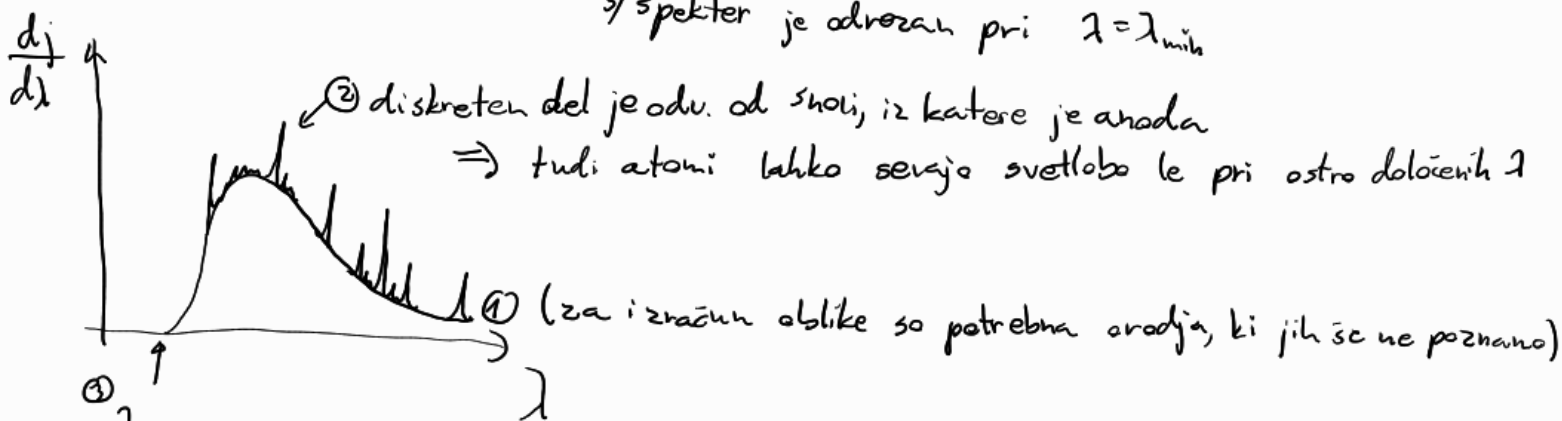
če vemo d , določimo λ in obratno



Presencenja:

- 1) gladek zvezni del spektra
(izvira iz 'klasičnega' zavornega sevanja)
- 2) na to je naložen diskreten del spektra ("vrhovi")

3) spekter je odrezan pri $\lambda = \lambda_{\min}$



① $\lambda_{\min}$

$\nu_{\max}$: ko e^- v anodi
pustijo vso svojo E_k ,
ki se pretvarja v
žarke λ

$$E_k = eU = h\nu_{\max} = h\frac{c}{\lambda_{\min}} \Rightarrow \lambda_{\min} = \frac{hc}{eU}$$

Comptonov pojav (Comp. sipanje)

- pojav, ki je kazal na kvantno naravo svetlobe

najprej: sipanje RTG žarkov na (skraj) prostih e^- v atomih

• pojav obravnavamo relativistično

PREJ:



ohranitev E: 1) $h\nu + mc^2 = h\nu' + \sqrt{p^2c^2 + m^2c^4}$

ohranitev p_x : 2) $h\nu = h\nu' \cos \theta + pc \cos \phi$

ohranitev p_y : 3) $0 = h\nu' \sin \theta + pc \sin \phi$

$$2)^2 + 3)^2: \quad p^2c^2(\underbrace{\cos^2\phi + \sin^2\phi}_1) = (h\nu - h\nu' \cos \theta)^2 + (h\nu' \sin \theta)^2$$
$$= (h\nu)^2 + (h\nu')^2 (\cos^2\theta + \sin^2\theta) - 2h\nu h\nu' \cos \theta$$

$$p^2c^2 = h^2(\nu - \nu')^2 + 2h^2\nu\nu'(1 - \cos \theta)$$

$$1)^2: \quad h^2(\nu - \nu')^2 + 2h(\nu - \nu')mc^2 + m^2c^4 = p^2c^2 + m^2c^4$$

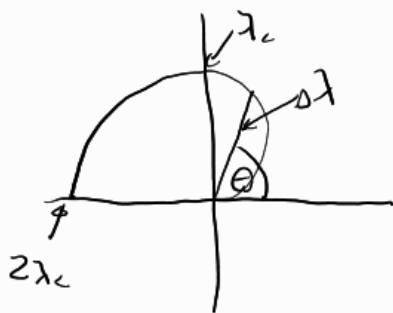
$$2h(\nu - \nu')mc^2 = h^2\nu\nu'(1 - \cos \theta)$$

$$\boxed{\lambda' - \lambda = \lambda_c(1 - \cos \theta)}$$

$$\Delta \lambda > 0$$

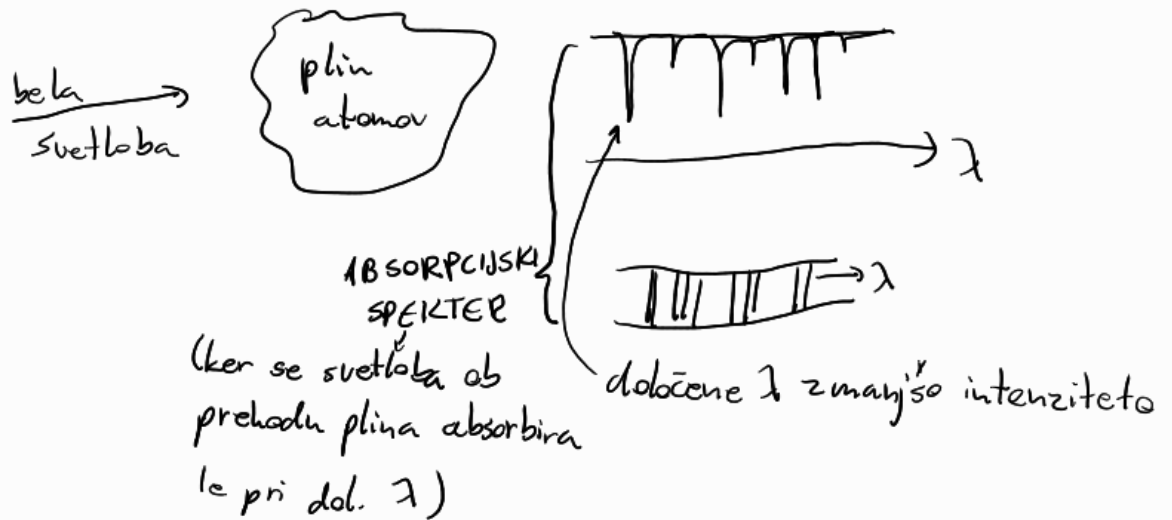
$$\lambda_c = \frac{h}{mc} \approx 2.4 \cdot 10^{-3} \text{ nm}$$

Comptonova val dolžina (za primer $X + e^-$ zelo majhna)

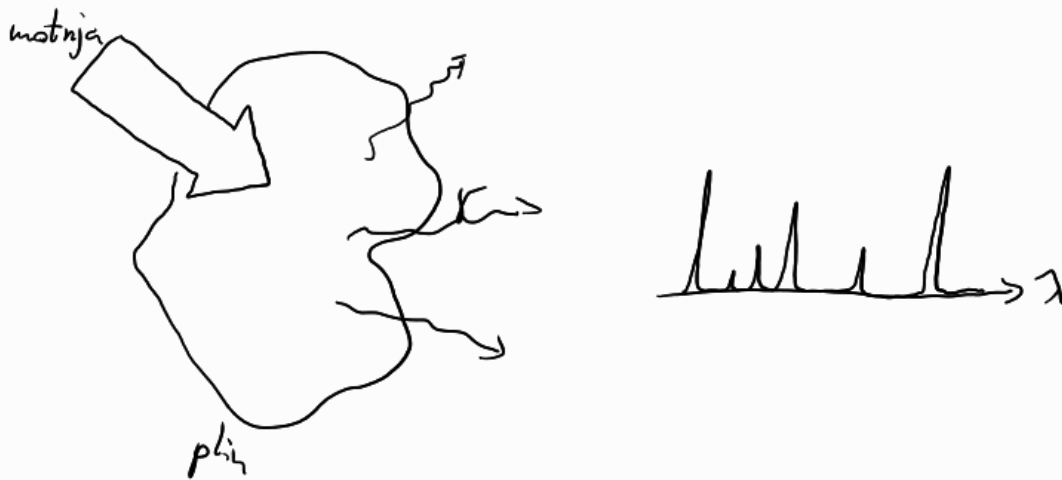


Atomski spektri (spektri sevanja atomov)

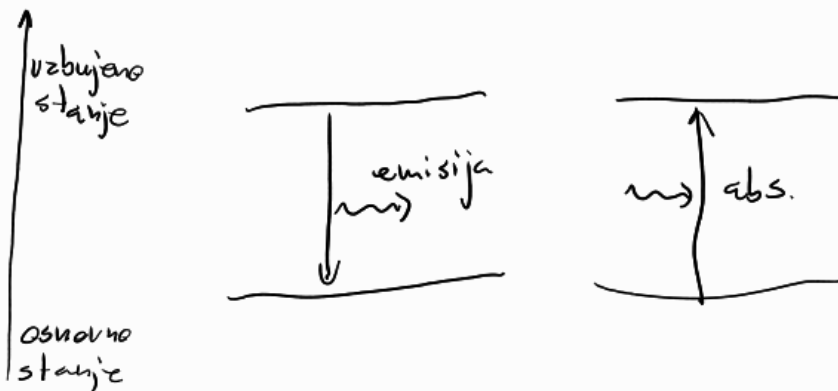
→ sevanja pri določenih λ ne opazimo le pri RTG sevanju, ampak tudi, ko atome vzbudimo in jih pustimo, da se de-ekscitirajo (ekscitirano)



→ če pa na plin atomov vplivamo z motnjo (sevanje, trki, ...), lahko tudi sam seva: EMISIJSKI SPEKTER

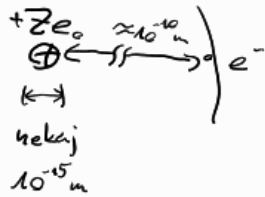


OSNOVNI MEHANIZEM



Bohrov model H-atoma

(ozadje: Rutherfordovo odkritje da je atom jedro + e^- oblak)



če bi imeli jedro $z +Ze_0$ (z protonov)

$$F_c = \frac{k Z e_0^2}{r^2} = \frac{m_e v^2}{r} \rightarrow v = \sqrt{\frac{k Z e_0^2}{m_e r}}$$

$$k = \frac{1}{4\pi\epsilon_0}$$

klasico naboj, ki kroži s frekvenco ν ,
seva svetlobo z isto frekvenco:

$$\nu = \frac{v}{2\pi r} = \frac{1}{2\pi r} \sqrt{\frac{k Z e_0^2}{m_e r}} \propto \frac{1}{r^{3/2}}$$

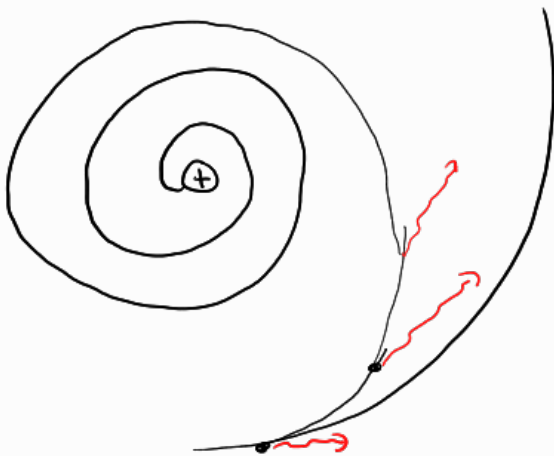
Celotna energija ($E_{\text{kin}} + E_{\text{pot}}$)

$$E = \frac{1}{2} m_e v^2 - \frac{k Z e_0^2}{r} =$$

$$= \frac{1}{2} m_e \frac{k Z e_0^2}{m_e r} - \frac{k Z e_0^2}{r} = -\frac{1}{2} \frac{k Z e_0^2}{r} \propto -\frac{1}{r}$$

$$(E_{\text{kin}} = -\frac{1}{2} E_{\text{pot}})$$

virialni teorem)



Klasično bi e^- "strmoglavil" v jedro
- to bi se zgodilo v $\approx 10^{-6} \text{ s}$.

$\Rightarrow$ atomi sploh nebi obstajali,
emisijski spekter nebi bil črtast, ampak
zvezen

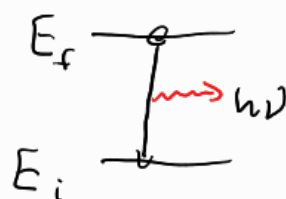
Bohr je težavo rešil z dvema postulatom:

- 1) e^- se giblje po določenih orbitah, ne da bi sevali (stacionarna stanja)
- 2) e^- seva ob prehodu iz enega stanja na nižje, energija izsevane svetlobe pa je dana po Plancku:
- 3) Korespondenčno načelo: ko kvantne preselimo v makroskopski svet (limita velikih kvantnih št.), moramo dobiti klasične rezultate

frekvenčni pogoj

$$h\nu = E_i - E_f$$

initial finite



Staz. stanja: kvantizacija vrtilne količine (VK):

$$L = m_e v r = n \hbar = n \frac{h}{2\pi} ; n = 1, 2, 3, \dots$$

↑
kvantno št. - označuje stanja

$$\Rightarrow r = \frac{n \hbar}{m_e v} = \frac{n \hbar}{m_e} \sqrt{\frac{m_e r}{k Z e^2}} \rightarrow r_n = \frac{n^2 \hbar^2}{m_e k Z e^2} = \frac{n^2 r_B}{Z} \propto \frac{1}{Z}$$

$$r_B = \frac{\hbar^2}{m_e k e^2} = \frac{4\pi \epsilon_0 \hbar^2}{m_e e^2} \approx 0,0529 \text{ nm}$$

Bohrov radij
(postavi prostorsko skalo atomskega sveta)

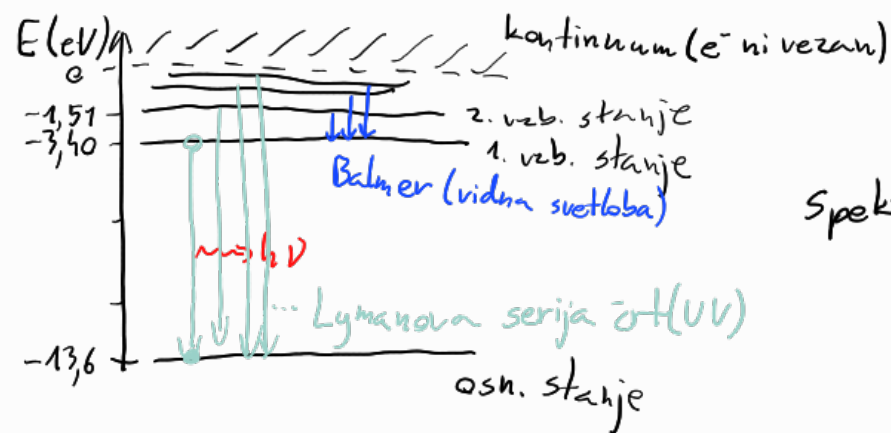
Celotna E e^- :

$$E_n = -\frac{k Z e^2}{2 r_n} = -\frac{m_e k^2 e^4 Z^2}{2 n^2 \hbar^2}$$

oz.

$$E_n = -E_0 \cdot \frac{Z^2}{n^2}, \quad E_0 = \frac{m_e k^2 e^4}{2 \hbar^2} = 13,6 \text{ eV}$$

Emissionijski spekter H atoma uzrno iz nivojske sheme:



Spekter:
$$h\nu = E_i - E_f = -E_0 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

$\Rightarrow E$ so kvantizirane

$\rightarrow$ vodi do (skoraj) pravega spektra

Ali velja korespondenčno načelo? (glejamo $n \rightarrow n-1$)

$\rightarrow$ kvantno:
$$h\nu = E_0 \left(\frac{1}{(n-1)^2} - \frac{1}{n^2} \right) = E_0 \frac{2n-1}{n^2(n-1)^2} \underset{n \gg 1}{\approx} E_0 \frac{2}{n^3} = \frac{m_e k^2 e_0^4}{h^2 n^3}$$

$\rightarrow$ klas.:
$$h\nu_{\text{klas.}} = h \frac{v}{2\pi r} = h \frac{n\hbar}{m_e r \cdot 2\pi r} = \frac{n\hbar^2}{m_e r^2} = \frac{m_e k^2 e_0^4}{h^2 n^3} \checkmark \text{ isto}$$

de Brogliejeva hipoteza (1924):

* svetloba se obnaša kot delci

de Broglie: ali se delci obnašajo kot valovanje?

} dualizen

$\rightarrow$ zapisal kot

$$\lambda = \frac{h}{p} \rightarrow \nu = \frac{E}{h}$$

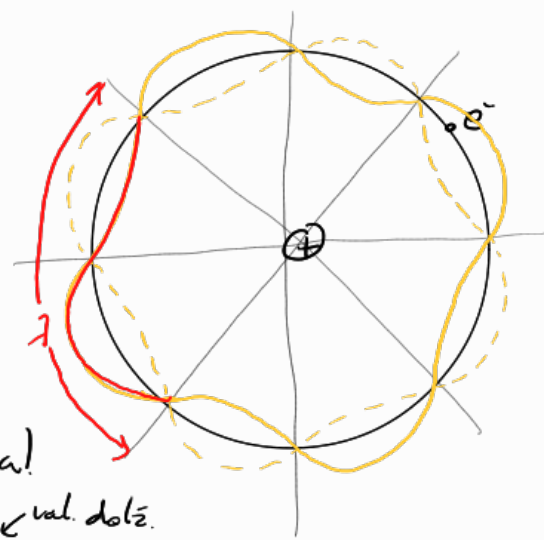
de Brogliejeva
val. dolžina
delca
GK delca

Za fotone je to avtomatsko
izpolnjeno:

$$E = pc = h\nu = h \frac{c}{\lambda}$$

$$v_k = m_e v r = n \hbar$$

$$0 = 2\pi r = \frac{n h}{m_e v} = \frac{n h}{p} = n \lambda$$



zakaj tega niso opazili že prej?

• λ za makroskopska telesa premajhna!

• uklonski pojavi znatni pri $\sin \theta \approx \frac{\lambda}{a}$
 uklonski kot $\leftarrow$ val. dolž. $\rightarrow \lambda \ll a \Rightarrow$ he vidimo nič
 velikost reže

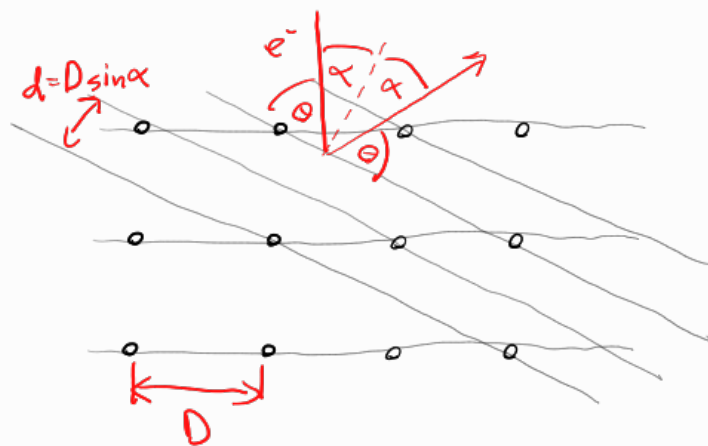
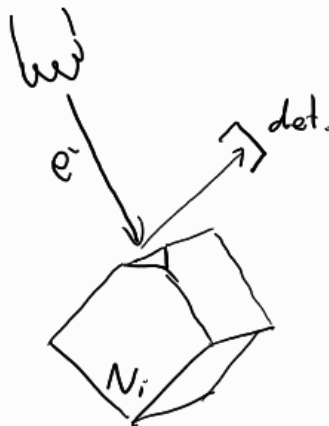
primer: ping-pong žagica: $m=2g$, $v=5 \text{ m/s} \rightarrow \lambda = \frac{h}{mv} = 6,6 \cdot 10^{-23} \text{ nm}$

$\rightarrow e^-$: $m_e = 9,1 \cdot 10^{-31} \text{ kg}$, $T=10 \text{ eV} = \frac{p^2}{2m_e} \rightarrow p = \sqrt{2m_e T} \rightarrow \lambda \approx 0,39 \text{ nm}$

primerljive a nudijo
 kristali (gradniki
 nekaj $0,1 \text{ nm}$ narazen)

Davisson - Germerjev poskus (1927)

("meritev valovne dolžine e^- ")



(Bragg) konstruktivna interferenca!

$$n\lambda = 2d \sin \theta = 2d \sin \alpha = 2D \sin \alpha \cos \alpha = D \sin 2\alpha = D \sin \varphi \leftarrow \text{merjeno}$$

Ni: $D=0,215 \text{ nm}$
 $\varphi=50^\circ$ ojačanje

$\rightarrow \lambda = D \sin \varphi = 0,165 \text{ nm}$, de Braglie $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_e e U}} \approx 0,167 \text{ nm}$ enako!

Opis "valovanja" delcev

Ravni val: $u(x,t) = A \cos(\omega t - kx)$, $k = \frac{2\pi}{\lambda} \overset{\text{de Broglie}}{=} \frac{2\pi p}{h} = \frac{p}{\hbar} \rightarrow \boxed{p = \hbar k}$

→ ne valovi delec, ampak verjetnostna amplituda oz. valovna funkcija:

$\Psi(x,t)$ (po Bornu)
podstavek
(ker je odv. od časa)

→ sama po sebi ni opazljiva, izmerimo pa lahko kompleksni kvadrat:

je opazljivo $\rho(x,t) = |\Psi(x,t)|^2 = \Psi^*(x,t) \Psi(x,t)$ verjetnostna gostota

Verjetnost, da delec najdemo (izmerimo) na dx : $dP = \rho dx$

$$\int_{-\infty}^{\infty} \rho dx = 1$$

(normalizacija verjetnostne gostote)

= delec zagotovo najdemo med $-\infty$ in ∞

- Če $\Psi(x,t) = A \cos(\dots) \Rightarrow \rho(x,t) = |A|^2 \cos^2(\dots)$



- Kaj pa $\Psi(x,t) = A e^{-i(\omega t - kx)}$?

$\Rightarrow \rho(x,t) = |A|^2 e^{i(\dots)} e^{-i(\dots)} = \text{konst.} \Rightarrow x$

Tak $\rho = |\Psi|^2$ ni videti kot delec, radi bi ga lokalizirali:

→ To naredimo z mešanico več k -jev:

npr. 2 k -ja: $k_1 = k + \Delta k$ in $\omega_1 = \omega + \Delta \omega$
 $k_2 = k - \Delta k$ $\omega_2 = \omega - \Delta \omega$

$\Psi_{1,2}(x,t) = \underbrace{\Psi_1(x,t) + \Psi_2(x,t)}_{\text{sešt. amplitude (kot v klas. fiz.)}} = A [e^{-i(\omega_1 t - k_1 x)} + e^{-i(\omega_2 t - k_2 x)}]$

$$= A \left[e^{-i(\omega t + \Delta \omega t - kx - \Delta kx)} + e^{-i(\omega t - \Delta \omega t - kx + \Delta kx)} \right] =$$

$$= A e^{-i(\omega t - kx)} \underbrace{\left[e^{-i(\Delta \omega t - \Delta kx)} + e^{i(\Delta \omega t - \Delta kx)} \right]}_{2 \cos(\Delta \omega t - \Delta kx)}$$

$$\rightarrow \rho_{1,2}(x,t) = |\Psi_{1,2}(x,t)|^2 = 4|A|^2 \cos^2(\Delta \omega t - \Delta kx) \quad \text{še vedno ni ok (N=2)}$$

Toda če ponovimo za $N=6$:

$$k_{1,2} = k \pm \frac{\Delta k}{3}; \quad k_{3,4} = k \pm \frac{2}{3} \Delta k; \quad k_{5,6} = k \pm \Delta k$$

dobimo sliko:



Ideja: $N \rightarrow \infty$

$$k_n = k \pm \frac{n \Delta k}{N}; \quad n = 1, 2, \dots, N$$

$$\omega_n = \omega \pm \frac{n \Delta \omega}{N}$$

$$\rightarrow \Psi(x,t) = A \lim_{N \rightarrow \infty} \sum_{n=-N}^N e^{-i(\omega t + \frac{n}{N} \Delta \omega t - kx - \frac{n}{N} \Delta kx)}$$

PREVEDEMO NA INTEGRAL:

$$\Psi(x,t) \approx N A e^{-i(\omega t - kx)} \int_{-1}^1 e^{-i u (\Delta \omega t - \Delta kx)} du = N A e^{-i(\omega t - kx)} \frac{\overbrace{e^{-i(\Delta \omega t - \Delta kx)} - e^{i(\Delta \omega t - \Delta kx)}}^{-2i \sin(\dots)}}{-i(\Delta \omega t - \Delta kx)}$$

$$\rightarrow \rho_{N \rightarrow \infty}(x,t) = |\Psi|^2 = \underbrace{4|A|^2 N^2}_{\substack{\text{ni važno} \\ (A \propto \frac{1}{N})}} \frac{\sin^2(\Delta \omega t - \Delta kx)}{(\Delta \omega t - \Delta kx)^2}$$



Približno izgleda kot delec, ki se
nan bliža kot val oz.
SUPERPOZICIJA VALOV (VALOVNI PAKET)

Gaussov valovni paket

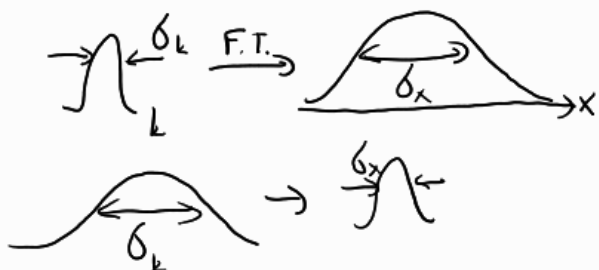
$\frac{\sin^2(\dots)}{(\dots)^2}$ paket: motijo nas oscilacije 

$$\psi(x) = \int_{\mathbb{R}} \underbrace{A(k)}_{\substack{\text{amplituda,} \\ \text{odv. od } k}} \underbrace{e^{ikx}}_{\text{ravninski val}} dk \quad (\text{Fourierova transformacija})$$

$$A(k) = A_0 e^{-\frac{(k-k_0)^2}{4\sigma_k^2}} \quad \text{Gaussovska odv.}$$

$$\begin{aligned} \rightarrow \psi(x) &= A_0 \int_{-\infty}^{\infty} \exp\left[-\frac{(k-k_0)^2}{4\sigma_k^2} + ikx\right] dk = A_0 \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2} \left(\frac{k-k_0}{\sqrt{2}\sigma_k} - \sqrt{2}i\sigma_k x\right)^2 - \sigma_k^2 x^2 + ik_0 x\right] dk \\ &= A_0 e^{-\sigma_k^2 x^2 + ik_0 x} \underbrace{\int_{-\infty}^{\infty} e^{-\frac{u^2}{2}} du}_{\sqrt{2\pi}} \cdot \sqrt{2}\sigma_k = \frac{1}{\sqrt{2}\sigma_k} \tilde{A}_0 \exp\left(-\frac{x^2}{4\sigma_x^2}\right) e^{ik_0 x}; \quad \sigma_x = \frac{1}{2\sigma_k} \end{aligned}$$

Začeli smo z $A(k) \propto e^{-k^2}$, dobili $\psi(x) \propto e^{-x^2}$ (Fourierova transformiranka Gaussove = gaussova.)
in $\sigma_x \propto \frac{1}{\sigma_k}$

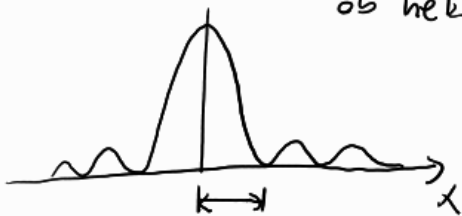


} bolj natančno kot določimo lege, manj natančna je njegova GK in obratno

Načelo nedoločenosti

lažje računati s paketom $\frac{\sin^2(\dots)}{(\dots)^2}$

ob nekem t



si izberemo kot nedoločenost lege

Ob $t=0$: $\sin \Delta k \Delta x = 0$

prvi minimum: $\Delta k \Delta x = \pi$

$$\Delta x = \frac{\pi}{\Delta k} = \frac{\hbar \Delta p}{2\pi \Delta p} = \frac{\hbar}{2\Delta p} \rightarrow \Delta x \Delta p = \frac{\hbar}{2}$$

oznake tudi $\delta_x \delta_k, \delta_x \delta_p, \dots$

produkt
nedoločenosti

⇒ Heisenberg:

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

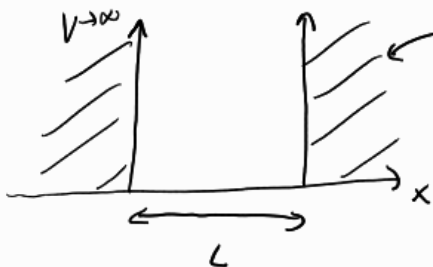
se da pokazati

(za Gaussov valovni paket
velja sp. meja $\frac{\hbar}{2}$)



ne moremo hkrati točno
določiti lege in GK

Zgled: delec zapremo v škatlo (1d, stranica L)



tu delec ne more biti

$$\Delta p = \overline{(p - \bar{p})^2} = \overline{p^2} \approx \left(\frac{\hbar}{\Delta x}\right)^2 \approx \frac{\hbar^2}{L^2}$$

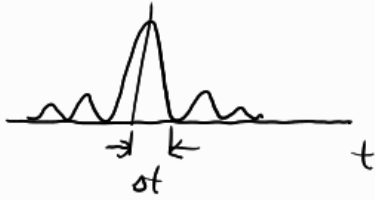
$$\begin{aligned} \sigma_x^2 &= \text{var}[x] = E[(x - E[x])^2] \\ &= \overline{(x - \bar{x})^2} = \dots = \overline{x^2} - \bar{x}^2 \end{aligned}$$

$$\rightarrow E_{\text{kin}} = \frac{\overline{p^2}}{2m} \approx \frac{\hbar^2}{2mL^2}$$

? Delec, zaprt v škatli, ima nen ničelno E_k !

→ ker smo delec lokalizirali ($E_p = 0$)

Analogno pogledamo



$$\sin \Delta \omega \Delta t = 0$$

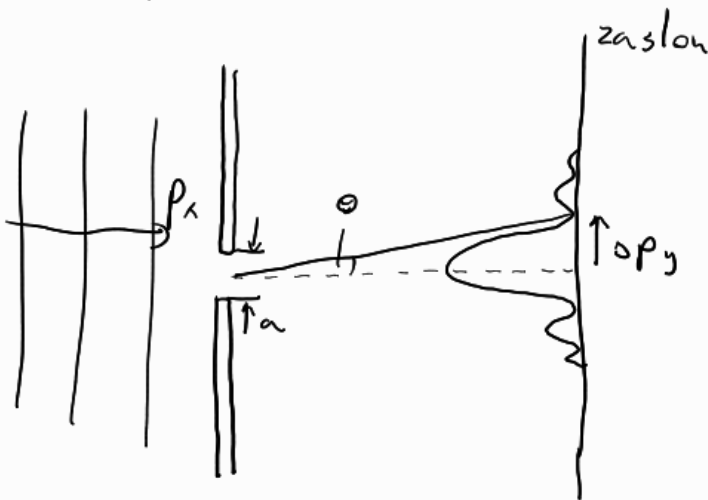
$$\Delta \omega \Delta t = \pi \rightarrow \Delta t = \frac{\pi}{\Delta \omega} = \frac{\pi \hbar}{\Delta E} \approx \frac{\hbar}{2 \Delta E} \Rightarrow \Delta t \Delta E = \frac{\hbar}{2}$$

$$\Delta E = \hbar \Delta \omega$$

$$\Rightarrow \boxed{\Delta t \Delta E \geq \frac{\hbar}{2}} \text{ Heisenberg za } t, E$$

Delec, ki živi dolgo, ima ostro določeno E

Negotovost pri uklonu na reži



$$p_y \approx 0 \pm \Delta p_y$$

pogoj za min.:

$$\underbrace{\frac{a}{\lambda} \sin \theta}_{\text{razlika poti žarkov}} = \frac{\lambda}{2}$$

$$\Delta p_y = p_x \tan \theta \approx p_x \sin \theta = p_x \frac{\lambda}{a} = p_x \frac{h}{p_x a} = \frac{h}{a}$$

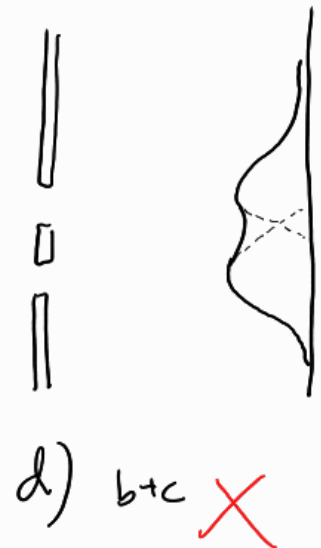
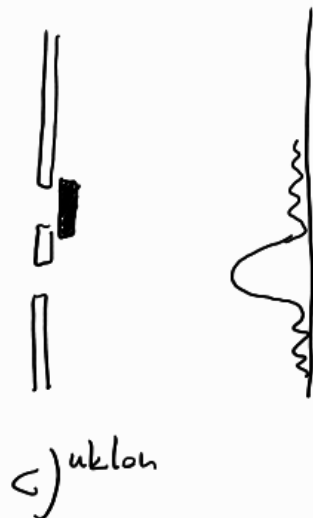
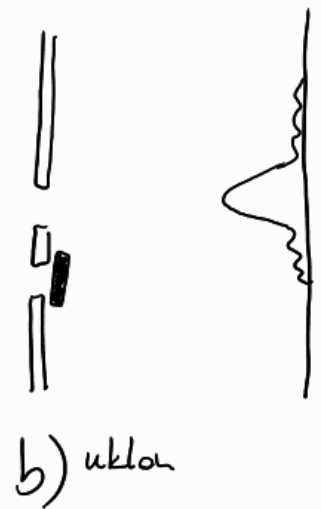
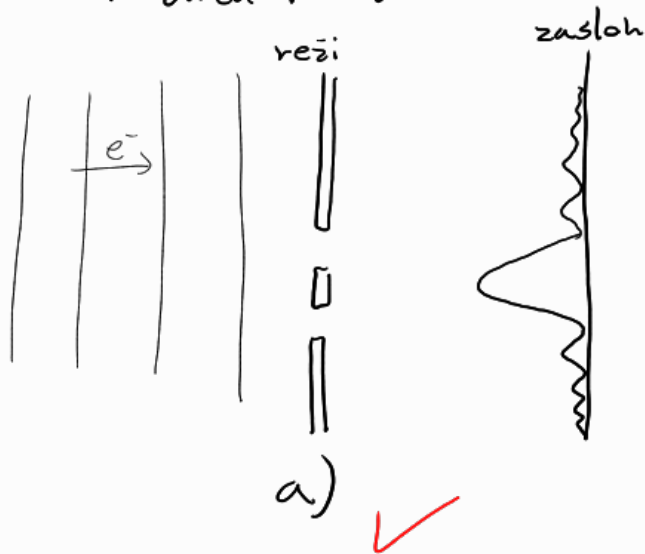
$$\Rightarrow \Delta p_y a = h$$

$$a = \Delta y$$

$$\Rightarrow \boxed{\Delta p_y \Delta y \geq \frac{\hbar}{2}} \text{ Heisenberg}$$

Interferenčni poskusi z delci

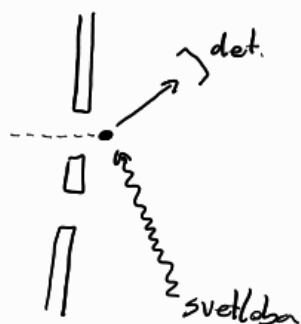
e^- na dveh režah:



Klasično bi pričakovali sliko d), v resnici se zgodi a).

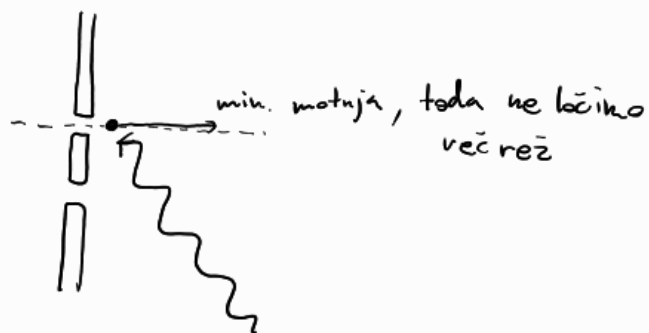
Tudi če pošljemo 1 e^- naenkrat, se zgodi a) $\Rightarrow$ e^- interferira sam s sabo

Skrozi katero režo gre delec?



- zagotovo identificiramo režo
- toda interf. slika izgine (meritev je zmotila e^-)

e^- manj zmoteno, če nanj posvetimo z večjo λ (manjša E)



ψ_1 = tir delca skozi režo 1

$$\psi_2 = -1 - 2$$

$$P_{\rightarrow} = |\psi_1 + \psi_2|^2 = |\psi_1|^2 + |\psi_2|^2 + \underbrace{\psi_1^* \psi_2 + \psi_1 \psi_2^*}_{\text{naredi interf. vzorec}}$$

$$P_{\rightarrow} = |\psi_1|^2$$

$$P_{\leftarrow} = |\psi_2|^2$$

$$P_{\rightarrow} = |\psi_1|^2 + |\psi_2|^2$$

↓
uporabimo, če ne moremo povedati, skozi katero režo je šel delec